Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Lesson 1: Sum Rule, Product Rule, and Bayes' Theorem

- Sum Rule, Product Rule, and Bayes' Theorem
  Reading • 15 min
- Sum Rule
  Reading • 4 min
- Product Rule (Chain Rule)
  Reading • 2 min
- Bayes' Theorem
  Reading • 43 min

Lesson 2: Gaussian Distribution

- Univariate Gaussian Distribution
  Reading • 4 min
- **Gaussian Distribution: Foundations and Applications**
  Reading • 15 min
- Multivariate Gaussian Distribution
  Reading • 10 min
- Conditional and Marginal Multivariate Gaussian Distributions
  Reading • 45 min
- Product of Gaussian Densities
  Reading • 26 min

Lesson 3: Inference Techniques

- Bayesian Inference
  Reading • 1h 10m
- Latent-Variable Models
  Reading • 50 min
- Check Your Knowledge: Inference Techniques
  Graded Assignment • 10 min

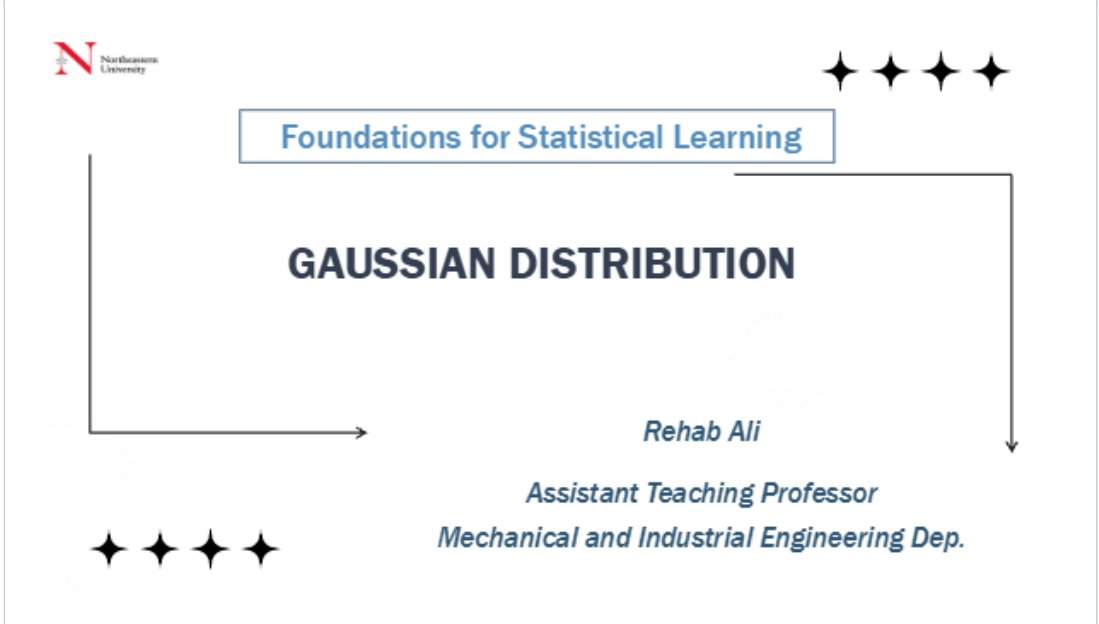
Module 4
Introduction to Model Evaluation and Optimization

Gaussian Distribution: Foundations and Applications

0:14 / 0:39 2x

Explore the following PDF for an overview of the Gaussian distribution, a fundamental concept in statistical learning. You will learn about its key properties, including its bell-shaped curve, and how it applies to both univariate and multivariate data. The document also covers important topics such as the standard normal distribution, conditional and marginal distributions, and the effects of linear and affine transformations on Gaussian random variables. This foundational knowledge is essential for understanding more advanced statistical and machine learning techniques.

1 / 15 100%



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